STACKS, DISCRIMINANTS: LEMMA-BASE-CHANGE-DIFFERENT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1812-1827 of sources/stacks-discriminant/pinned-original-source.tex; proof span is lines 1829-1860, the complete author proof (30 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 15 of queue_discriminant.json (pb=1456, stmt 1812-1827, proof 1829-1860).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 07T8: sources/stacks-discriminant/pinned-original-source.tex, source lines 1860–1860. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BUR: sources/stacks-discriminant/pinned-original-source.tex, source lines 1860–1860. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00L2: sources/stacks-discriminant/pinned-original-source.tex, source lines 1860–1860. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BVF: sources/stacks-discriminant/pinned-original-source.tex, source lines 590–593. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BT9: sources/stacks-discriminant/pinned-original-source.tex, source lines 894–907. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:discriminant.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Consider a cartesian diagram of Noetherian schemes*

$$\begin{array}{ccc} Y' & \longrightarrow & Y \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ X' & \xrightarrow{g} & X \end{array}$$

with f flat and quasi-finite. Let $R \subset Y$, resp. $R' \subset Y'$ be the closed subscheme cut out by the different $\mathfrak{D}_f$, resp. $\mathfrak{D}_{f'}$. Then $Y' \to Y$ induces a bijective closed immersion $R' \to R \times_Y Y'$. If g is flat or if $\omega_{Y/X}$ is invertible, then $R' = R \times_Y Y'$.

Proof. There is an immediate reduction to the case where X, X', Y, Y' are affine. In other words, we have a cocartesian diagram of Noetherian rings

$$\begin{array}{ccc} B' & \longleftarrow & B \\ \uparrow & & \uparrow \\ A' & \longleftarrow & A \end{array}$$

with $A \to B$ flat and quasi-finite. The base change map $\omega_{B/A} \otimes_B B' \to \omega_{B'/A'}$ is an isomorphism (Lemma 0BVF) and maps the trace element $\tau_{B/A}$ to the trace element $\tau_{B'/A'}$ (Lemma 0BT9). Hence the finite B-module $Q = \mathrm{Coker}(\tau_{B/A} : B \to \omega_{B/A})$ satisfies $Q \otimes_B B' = \mathrm{Coker}(\tau_{B'/A'} : B' \to \omega_{B'/A'})$. Thus $\mathfrak{D}_{B/A}B' \subset \mathfrak{D}_{B'/A'}$ which means we obtain the closed immersion $R' \to R \times_Y Y'$. Since $R = \mathrm{Supp}(Q)$ and $R' = \mathrm{Supp}(Q \otimes_B B')$ (Algebra, Lemma 00L2) we see that $R' \to R \times_Y Y'$ is bijective by Algebra, Lemma 0BUR. The equality $\mathfrak{D}_{B/A}B' = \mathfrak{D}_{B'/A'}$ holds if $B \to B'$ is flat, e.g., if $A \to A'$ is flat, see Algebra, Lemma 07T8. Finally, if $\omega_{B/A}$ is invertible, then we can localize and assume $\omega_{B/A} = B\lambda$. Writing $\tau_{B/A} = b\lambda$ we see that $Q = B/bB$ and $\mathfrak{D}_{B/A} = bB$. The same reasoning over B' gives $\mathfrak{D}_{B'/A'} = bB'$ and the lemma is proved. □